

project_ASR_stutter master document

Key

Black: Patient

Blue: Second speaker

Green: Third speaker

Red: Grave errors in output

(Text in parentheses) or _{subscript text}: Words that are said quietly and may not count towards the total word error rate if the output does not match in this area (upon further discussion)

Substitutions that will not be penalized (will not count towards total word error rate)

- These words will be kept in the transcript, so if the output matches any of these forms, it will still be considered correct.
- Not all substitutions fall under this category, only those deemed to be reasonably close.

Word(s)	Output
Ain't	Isn't
Don't, as a contraction for doesn't	Doesn't
Any gerund contraction (ex. doin')	Its uncontracted equivalent (ex. doing)
Word forms of numbers (ex. twenty)	Number forms of the word (ex. 20)
Number forms	Word forms of the number (must be exact number as the original, so thirty does not count as a reasonable substitution for 20)

- Outputs that evidently do not meet the word error rate requirement of 1% may simply not be evaluated for word error rate

What will be removed from both the transcript and the output of any models

- Filler words such as "um", "uh", or "mm hmm"

Wernicke's

1. (medium quality)

<https://www.youtube.com/watch?v=aVhYN7NTIKU>

Black: patient

Blue: interviewer

They're good, you know. Both. **What's this?** You know, the. Teeth. and right there. I. End of the football. **End of the football?** Yes. As uh, uh, uh, and so it's a football. Right here. like, Cause it to comes in. See? **Yeah.** The end of it, see. Oh, we have all of didn't see. **I don't understand.** Well, it's uh, well it's quite a pile of the discovery, see. You don't understand it, but it applies in that long thread of each comin' from that way, and it comes starts in the centi out here, see. Oh, young good to see you, thumbs up, and finally come out here and goes out to you. Get it? **Sort of.** Yes, it comes out here, see. Comes out here, here, you. And and here. **Uh huh.** And it comes here, and it comes out, and bing. That's a good spectrum, see. **That's a good what?** That's a good foul throw both here. Fly, flyer, see it flows cold or gives or kicks out or bing, whatever it is, it we has no fit football fired out, it's just pulley, see, but they say feel them, uh, I guess you're meant to have beds, see I don't know. [chuckle]

Duration of transcription: beginning to 01:28.1

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They're a foot, you know, boom. **What's this?** You know, there's big anger right there. All right. End of the football. End of the football? Yes. Is it, is it, is it, and so it's a football right here. Like, cause it comes in, see? **Yeah.** The end of it, see. Oh, we have other things, see. I don't understand. Well, it's quite a parable in the other, see. You don't understand it, but it's probably just in the long, sort of, even comes from there. And it comes source in the country, out through, see. All the under and see, it comes up. And finally, come out here and goes out, see. You have it? **Sort of.** You know, it comes out here, see? It comes out here, here, and here. And it comes in, and it comes out, and being. That's a good fastball, see? That's a good what? That's a good, good flow of, flow of those, fire, fire, see? It flows cold, or it keeps it kicks out, or being, or whatever it is. We have no football, no doubt. It's just pulling, see? But they say, feel those. Oh, I guess you meant you have bets, see? I don't know.

small.en

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large-v1

Gray represents (un)educated guesses made from unclear audio
Cut all clips to under 90 seconds to reduce software workload

2. (1:03 long clip) <https://www.youtube.com/watch?v=stdjJeU1doI>

Black: Primo (patient)

Blue: Dr. Lori (interviewer)

So if we look at a calendar in general, Mm hmm. how often do you guys get together? To make. Like maybe once a quarter, To make an es, save of base, starting as one, two, three... Okay. And fifth in, October, November, December. Six, six of them. Okay, so there's only six meetings. Also, also velt, but they have a, a uh, cella, a, favela lavir, uh uh, fabir, famigus. So you told me like March, April, May, and then October, November, December? Yeah, in the uh, in the... So is it for snowbirds? Yes, you do, yes... Okay. So maybe six months out of the year total? Um, memen Mondays we ways yes a [...] Okay, with that bin? I'll get you a different one.

base.en

So if we look at a calendar in general, how often do you guys get together? They make you, maybe once a quarter. They make a stave of a's, starting as one, two, three. Okay. And 15 October, November, December, six, six of them. Okay, so there's only six meetings. Also, also, but to have a, a stellar, for, for beer, for Vegas. So you told me like March, April, May, and then October, November, December. Yeah, in the, in the, in the. So is it for snowbirds? Yes, yes, yes, yes. Okay. So maybe six months out of the year total. And, and, and, one of the sweet ways, yes, so. Okay. With that pen, I'll get you to do this.

small.en

medium.en

large-v1

3. <https://www.youtube.com/watch?v=cJ6OvHq9WrQ> (8.901 to 1:53.400 creating a 1:44.499 long segment)

Black: Jim, the husband and patient

Blue: Amy, the wife

Green: I don't know who that is, but she gives the interview

Wire, wire begs. [Oh, are you talking about whitewater? Yes, yes.] When we were kids, down near bellends and seleybins and we weres, wiffen wellerhens. Jefferhan jefferhans. We were kids. Is it whitewater? [No no, no. Not Six Flags.] Refordhans. is, is exactly right there. (relley) Relley friends... Sandy Springs. It's this kind of, it was it was a, it was a mars, it was a marsons, a marshen, a marsons. It was a, something with [lots of water? Yes, yes.] I don't know. Yeah but just, it's it's. I didn't grow up, that where You were, you with me(ds), and, when we were kids, we went all the way down to here, Chattahoochee? That's going to be a long one to write. [(writing it down) Chattahoochee. Cause I'm just kinda confident.] Yeah, that, that, that is where I used to camp, but you, you and me [...] So it's a place you and Amy [went... Yeah,] I mean literally, literally, did of, there's one, What happened at that place? Give [us more clues. Like what did we do there?] So you were kids, and what else? What did we do there? We, (be) before the kids, probably without our little kids, so that's how many years (we pr) we probably younger were on plays the lights. And., [and, Where did we] food, the food, is the (k) there's a, there's a, (it's) it's a, it's like a... It's a kind of food? Yes, yeah, it's a tysa place... So this is all about food? This is all about food.

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wire wire bags oh yes yes we were kids down near villains and selling bins and we were with weller hands Jeffra hands Jeffra hands we were kids white water no not six legs refered hands is exactly right there's really really friends sandy springs it's this kind of it was a morse it was a Marson's a Marson it was a something with water I don't know yeah but just I didn't grow up that way here at us me I when we were kids we went all way down in here I'm just kind of confidence yeah that's that's that is where I used to count us but you you and me so it's a place you and you went I I mean little it lonely that there's one what happened to that place give us what we do there so you were kids and what else what do we do there would be before the kids probably without our little kids so that's how many years we probably younger one on plays the lights and where the food the food is the there's it it's it's a it's like it's like a it's food yes yes it's a tighter place

Analysis: What happened here makes no sense. It appears it's only transcribing what Jim says, identifying him as the main speaker and others as background noise, but that is not what we want it to do.

small.en

medium.en

large-v1

unused

4. <https://www.youtube.com/watch?v=B1SBFUgluK0>
5. <https://www.youtube.com/watch?v=AqcwPqDaYnw> (unused, probably not actual aphasia)

Broca's

1. <https://www.youtube.com/watch?v=RMa9BVpJkYQ>

(Most of this video isn't the patient speaking, but this represents a real interaction of someone with Broca's aphasia, so I will use it.)

Black: Kent (patient)

Blue: interviewer

elisions (like doin' for doing) are represented in full

Yes. So how are you doing today, Kent? s... Tim set of... tubers... okay. Okay? Are you having pain anywhere? No. You feel pretty good in general? Yeah yes. You can understand what I'm saying to you, right? Yes. Okay. Can you close your eyes for me? Perfect. Now tell me your name. George t... cause I've h... I've had... No... I am... I've had... Can you tell me what city we're in right now? I bad. This is Richmond. Can you say Richmond? Yes. Rich of... Richmond. Rich is good. Can you say one two three? One two three. Okay. That's very good. Can you name the weeks of the uh, days of the week for me? yes. Name the days of the week. I am I...I have had. I. (simultaneously) [As far as you can do. Sunday.] Susan. Fancy. Thirty. Friday. Saturday. Sunday. Okay. That's good. What day is it today? Monday. Monday... Can you tell me what the weather's like outside? Yes. Um... Seeder. Yes. Frustrating, ain't it? yes. Okay, thank you very much. Thank you.

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Yes. So how are you doing today, Kim? Timb-sitter. Timb-s. Okay. Okay. Are you having pain anywhere? No. You feel pretty good in general? Yes. You can understand what I'm saying to you,

right? Yes. Okay. Can you close your eyes for me? Perfect. Now tell me your name. Yes. Go... Because I... I'm bad. I'm not gonna try to... No. I'm bad. I'm bad. Can you tell me what city we're in right now? I'm bad. This is Richmond. Can you say Richmond? Yes. Richmond. Richmond. Richmond. Can you say one, two, three? One, two, three. Okay. That's very good. Can you name the weeks of the... Days of the Week for me? Yes. Name the days of the week. I'm... I'm... Sorry. Tuesday. Fancy. Firty. Firty. Saturday. Sunday. Okay. That's good. What day is it today? Monday. Monday. Yes. Can you tell me what the weather's like outside? Yes. Tuesday. Yes. Frustrating unit. Yes. Okay. Thank you very much. Thank you.

What it tends to do more: Tends to hypercorrect words more than usual, assumes things not inherent in speech (says Tuesday near the end when it doesn't sound like "Tuesday")

There were sections where both of us failed (near the beginning when the speaker failed to sound anything out and both I and the base model has to guess)

small.en

medium.en

large-v1

2. <https://www.youtube.com/watch?v=f2IiMEbMnPM> (time: until 2:24.823)

Now tell me what this thing was with your legs last week or week before. Uh, no good. Uh... uh, ache, ... and... uh, ... uh, uh ... knees, and ankles, uh ... What'd you do about it? Uh, home, ... doctor, and, legs, uh, walking, no good, [uh... No good, huh?] And you're a fella who likes to [walk, too. Oh, geez.] yeah, huh? Oh boy, oh boy. And it's getting too cold, Yeah, to run your John Deere tractor. [yeah, Ain't it?] yeah. Up and down town. Yeah. What'd the doctor tell you, Jack? Don't know. He don't know what's going on? No, no. Did you feel sick in any other way? No. No, no. Uh, ... uh, ache. [Legs ached? Uh,] Yeah. Uh... Did you feel weak? No. No, no. No, uh... Maybe had to touch a room a tis. Don't know, gee. Uh...

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Now tell me what this thing was with your legs last week or week before. No good. Oh. Oh. Take. And. Oh. Oh. Oh. Oh. Knees and ankles. Oh. What'd you do about it? Oh. Home. Doctor. And legs. Oh. Walking. No good. And you're a fellow who likes to walk. Oh, Jesus. Oh boy, oh boy. And it's getting too cold. Yeah. To run your John Deere tractor. Yeah. Up and down tail. Yeah. What'd the doctor tell you, Jack? Don't know. He doesn't know what's going on. No.

Did you feel sick in any other way? No. No. No. A. Yeah. Legs ached. Yeah. Did you feel weak? No. No. No. No. Maybe you had a touch of rheumatism. Don't know what I did. Oh.

small.en

medium.en

large-v1

3. (UK English) <https://www.youtube.com/watch?v=IP8hkopObvs> (time: 1:29.211)
So what's your name? Um, Scott. Oh, no. Sarah Scott. That's right. And how old are you? I can't. Try. I can't. You're nineteen. Nineteen. Yeah. And what happened to you? Um, stroke. Um, school. And English class. Okay. And I, um, book, and I read it aloud. But I can't, cause, stroke. And what, ... happened after the stroke with your speech? Well I, I can't. What do you do now, every day? Oh, um, ... speech, lang, um, the speech. Therapy. Yeah. Okay. And, um, also, riding, um, Riding what? Oh, um, ... horses. Yeah, good. And how do you feel? I'm fine. You're fine? That's good.

base.en

So what's your name? Um... Scott. Oh. No. Sarah Scott. That's right. And how old are you? I can't. Try. I can't. You're 19. 19. Yeah. Yeah. And what happened to you? Um... stroke, um... school, and English, class. Okay. And I, um... book, and I read it aloud. Um... but I can't, cause... stroke. And what... happened after the stroke with your speech? Well, I... I can't. What do you do now? Every day. Oh, um... speech, then... um... the... speech. Therapy. Yeah. Okay. Um... also... writing. Um... writing what? Oh. Um... horses. Yeah, good. And how do you feel? I'm fine. I'm fine. Yeah, fine. It's good.

small.en

medium.en

large-v1

Conduction aphasia (not either type)

fluent speech and comprehension, but unable to repeat words or sentences

1. <https://www.youtube.com/watch?v=G94TvTvjeeU> (time: 0:00 to 0:41.098)
The producer asked us to count to 10. Remember that? Yes. Can you do that again? Why? One, two, three, four, five, seven, n, boyn, two, gebrys, people, go I can't [no, I, No,] That's really hard,

let me get you started. I was stunned, *okay*. [One, One,] Two, four, five, six, bedder, send, poind, uh, *Good enough*. [Let's... Is that all right?] *That's fine*.

base.en

The producer asked us to count to ten. Remember that? Yes. Can you do that again? Why? one two three four five seven Boy to Gabriel people Go I can't know that's really hard. Let me get you started. I was done. Okay one one two four five six Bander send point Good enough, let's go right. That's fine

small.en

medium.en

large-v1

2. <https://www.youtube.com/watch?v=N91JMnoVSfl> (this one probably isn't because he repeats the same word over and over again, but it was classified under conduction aphasia and was one of the few I could find)

This person isn't actually saying the word tunnel, but just a repeated phrase that sounds like it. *Now tell me, why is it that you're moving, actually?* Uh, tunnel tunnel tunnel tunnel tunnel tunnel tunnel, tunnel tunnel tunnel tunnel tunnel tunnel tunnel tun... uh tunnel tunnel tunnel tunnel tunnel tunnel tunnel tunnel, tunnel tuh, uh tunnel tunnel tunnel tunnel tunnel, uttel tunnel tunnel tunnel... *Is it because the kids have grown, and the house is [too big?* Tunnel. Tunnel tunnel tun, tun, tunnel tunnel tun tun... *Ready for a smaller place?* Tunnel. *Seems reasonable. Must to keep up.* Ton tunnel tun. A tunnel tunnel tunnel tunnel tunnel, a tunnel, a ton ton ton ton tunnel, tunnel tunnel ton. Ton ton tunnel. *All your children are grown, right?* Oh tunnel tunnel. *How many children do you have?* Ton ton tunnel. Ton ton... *Can you count that out?* Tunnel tunnel tunnel tunnel tunnel ton, a tunnel ton, tunnel tunnel tunnel tunnel tunnel toll, [t]unnel tunnel tunnel, a tunnel tunnel tunnel tunnel tunnel ton. Don't, ton ton. Ton tunnel tunnel tunnel tunnel. ... Tunnel tunnel tunnel *What number is this?* Duntel. *Right? Now let's count it out.* One, two, three, four. *Nice, nice. Well since you did four so well, let's go for broke. Let's go up to twenty if you can.* A toll. One, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, teen, tunnel, tunnel, ... One, to tunnel tunnel tunnel tunnel to tunnel ton. *Okay*.

[base.en](#)

small.en

medium.en

large-v1

3. <https://www.youtube.com/watch?v=34kj3oqzZks>

unused

https://www.youtube.com/watch?v=1yC-uX5VOiE&list=PLVEF_12Z5MxtIf5P4u9MnvMYmpZR1_h5V&index=6 (this one I chose not to transcribe because it appears the "patient" is just a paid actor because she appears in many other videos of different types of aphasia with different severities)

Anomic aphasia (none of the above types)

Characterized by inability to find specific words for things (is milder than Broca's or Wernicke's)

1. <https://www.youtube.com/watch?v=w95EF3fW2IA> (from 01:01.681 of the original to ???)

Your first object is this. Can you think of the name for that? No. Okay. So, first, what shape is it? It's round. Why don't you write down underneath that it's round? Okay. Where could you find it? Uh, sometimes it's in my kitchen? Yeah, you could find it in your kitchen, so why don't you write down kitchen? ... Who could use it? Um, I could use it? I like to eat it sometimes. Yeah, you eat it, so actually, you can write that right here, that you eat it. What does it look like, Catherine? It's um, it's orange. It's orange. What else? Why don't you feel it? What's it feel like? It's hard. It's hard. Okay. Can you smell anything? It smells sweet. Does it smell familiar? Have you smelled that before? I think so. Okay. Why don't you write down some of the things that you just told me? Okay, so look at everything that you've told me. You've told me it's round, it's hard, it smells sweet, you find it in your kitchen, you can eat it... Can you think of the word for it, Catherine? Keep looking at all of your clues. Is it an orange? It is. It's an orange. Say it again. Orange. Write it down. ... What is it? It's an ... orange. Here, hold it and say it again. It's an orange. Good job.

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Your first object is this. Can you think of the name for that? No. Okay. So first, what shape is it? It's round. It's round. Okay. Why don't you write down underneath that it's round? Okay. Where could you find it? Sometimes it's in my kitchen. Yeah. You could find it in your kitchen. Okay. So why don't you write down kitchen? Who could use it? Um, I could use it. I like to eat it sometimes. Yeah. You eat it. So actually, you can write that right here that you eat it. What does it look like, Catherine? It's um... It's orange. It's orange. What else? It's orange. Why don't you feel it? What's it feel like? It's hard. It's hard. Okay. Can you smell anything? It smells sweet. Does it smell familiar? Have you smelled that before? I think so. Okay. Why don't you write down some of the things you just told me. Okay. So look at everything that you've told me. You've told me it's round. It's hard. It's smell sweet. You find it in your kitchen. You can eat it. Can you think of the word for it, Catherine? Keep looking at all of your clues. Okay. So what's the name of the word? It's a word. It's a word. It's a word. It's a word. It's a word. It's a word. It's a word. It's a word. It's a word. It's a word. Just kidding. But we're having rightATION... So say it again. It's a word I like to make most of our sure children that teach them. See?

small.en

medium.en

large-v1

2. <https://www.youtube.com/watch?v=ynqaAShHRQo> (from 10.931 to end)

Okay, so this is your object right here, so why don't you tell me the shape of it? It is ... a rectangle? Yeah, it's a rectangle, why don't you write that down? Okay, so where could you find it? Well, my kids used to use these at school. Yeah, so maybe you could find it in school, in a classroom, or with your kids in school. So you can write all that down. Okay. What does it do? It um, it holds your, it holds your papers? Okay, why don't you write that one down. What does it look like? That one's kinda hard, so why don't you play with it for a while and figure it out. It's ... kind of stiff, and it has these pockets in it? Good! Okay, so why don't you look at everything you wrote down. Oh, I know what it is! It's a folder. Very good, Catherine! It's a folder. So you've used these with your kids before, right? Mm hmm. Good, and you send homework home in it.

Yeah. *So what's it called?* It's a folder. *Good.* Here, hold it again. *Why don't you write it down?* Folder. *Good job, Catherine.*

3. <https://www.youtube.com/watch?v=GFar6xGeZ9k> (from 1:54.900 to 3:20.662)

The first one is... Okay. Green group, it is in electronic? *Okay.* Um, blue, do, I can type on it. *Mm hmm.* And I can use it to send emails. Um, white, what does it look like, it is, rectangular, and um, it, I think this is the classic? What is it made of? It's made of keyboard, and plastic and ... *What about this part, Catherine?* This is a mouse, so that would be the part you use? *Right.* White, where? I see it at school and I think we have one of these at home. Um. *The last one is just what else do you know about it. Have you used this before?* Yeah. *Where?* Um, at school, and at home, when I'm writing notes. *Good, so have you written notes on it at school when you were with your students?* Mm hmm. *Okay, so you know what it is now and you're going to do all those things, can you think of the name for this?* I don't know.

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The first one is... Okay, green group, it is an electronic. Okay. Blue, do. I can take on it and I can use it to send emails. White, what does it look like? It is recting you. And I think it's plastic. What is it made of? It's made of keyboard and plastic. What about this part, Catherine? This is a mouse. So that would be the part. Can you use white? I see it at school and I think we have one of these at home. The last one is just what else do you know about it? Have you used this before? Yeah, at school and at home when I'm writing notes. Good, so have you written notes on it at school when you were with your students? Okay, so you know what it is now and you go through all those things. Can you think of the name for this? I don't know.

small.en

medium.en

large-v1

4. <https://www.youtube.com/watch?v=58SZBJxlCM>

Notes (Other)

- These people often cannot speak coherently because of brain damage
- These models can help make their speech coherent using context
- interesting frontiers for speech recognition]

- use more sophisticated LLM for doing transcription

(Phase 2, after this is finished: find clips of ethnic dialects of English or other types of dialects, not necessarily with aphasia)

compare this to a transcription from ChatGPT, do prompt engineering,

- Start with British English, Australian, NZ, Canadian English
- Move on to AAVE, Indian, South African English, Singaporean English